Phase 5 — Execution · phase5_execution/ (Paket: execution)

Erster Order-Pfad: Phase-4-Candidate → Gates → echte (IG Demo) Order. Manueller Trigger, KEIN Scheduler (= Phase 8).

✓ code-complete + live-verified · ✓ 130 unit tests · ✓ live_test 4/4 (Demo) · ⚠ live-verified ≠ profit-validated

EINGÄNGE

- **IG Broker (ig_demo)** — get_price · get_ohlcv · positions
- **turbo_candidates.json · Phase 4** — frozen 10-Feld-Kontrakt (direction ko
- **Database · Phase 2** — get_risk_level() (READ-ONLY)
- **ExecutionConfig (frozen)** — Fenster · VETO-Schwellen · Sizing · SL/

Session-Health · _ensure_session()
is_connected / connect / get_account.ok → sonst ExecutionAbort (1× pro Lauf, reused)

Reconcile-on-Startup · Decision E
reconcile_positions(expected_refs): missing→mark_closed · unexpected→Abort (fail-closed)

Gate 1 — Trading-Fenster
09:00–17:30 Europe/Berlin (konfigurierbar, tz-aware now_fn) → außerhalb = NO_TRADE

Gate 2 — Candidates laden ◆ AI
load_candidates / candidates_are_fresh · stale → lazy research_runner() · leer = NO_TRADE

Gate 3 — Constraints
Budget > 0 · max_parallel

Gate 5 — Direction
Konsistenz, KEIN FLIP

(teilen 1× get_open_positions) · Gate 5 = gate_direction_consistency

Gate 4 — Sizing · Decision F (⟳ reworked 2026-06-12)
Risk-per-Trade ÷ Stop-Distanz: size = balance·risk_pct / (stop_pts·pt_value)
Notional-Cap max_leverage · ab auf 0.1 · < min_deal_size → NO_TRADE

pre_trade_check — 4 HARTE VETOs
frischer Snapshot unmittelbar vor Order · erster Fail = kein Trade · Datenfehler → fa

1 · Status / Zeit (TRADEABLE) | 2 · Spread frisch ≤ max
3 · Momentum ◆ get_ohlcv | 4 · Position-Conflict

build_order_plan
deal_reference bot-… · absolute SL/TP-Level (BUY: stop unter/limit über) · currency (EUR)

Human-Confirm · Decision D
require_confirm default AN · --yes Override · abgelehnt = ABORTED_BY_USER (keine Order)

place_order · Decision E (Idempotenz)
record_pending WRITE-AHEAD → open_position(deal_reference, currency)
ACCEPTED→mark_open · PENDING/UNKNOWN→bounded re-check, sonst fail-closed (NIE 2.

monitor_position · Decision H (Polling)
Loop poll_interval_s: Position weg → CLOSED_BY_BROKER (broker SL/TP gefüllt)
square_off / max_hold → close_position → TIME_STOP · now_fn/sleep_fn injiziert

AI-GRENZE
„So wenig AI wie möglich…"
Phase 5 = KEINE AI.
Einzige AI: lazy P4-research_runner (Gate 2)

↪ NAHT FÜR PHASE 6
Bull/Bear/Judge-Debatte ersetzt die Richtungs-/Confidence-Quelle hier — OHNE Gates neu zu schreiben.

Nahtstellen:
- gate_direction_consistency
- build_order_plan

VETOs sind HART & AI-blind
Phase 6 darf KEINEN VETO überschreiben (kein Force-Trigger). Debatte liefert nur Richtung; Gating bleibt deterministischer Code.

LIVE-BEWEIS (2026-06-12)
SELL Demo-Order ACCEPTED
deal_id=DIAAAAXQ6CSGTA5
→ monitor → Time-Stop-Close
→ Record CLOSED · RESULT 4/4

VETO 3 live bestätigt:
BUY into drop -0.20% → veto
Schwellen = v1 (Phase 7 tunen)

done / live-verified · harter VETO (no trade) · AI / Phase-6-Naht · Logging → stderr · Result-JSON → stdout · editable installs (kein